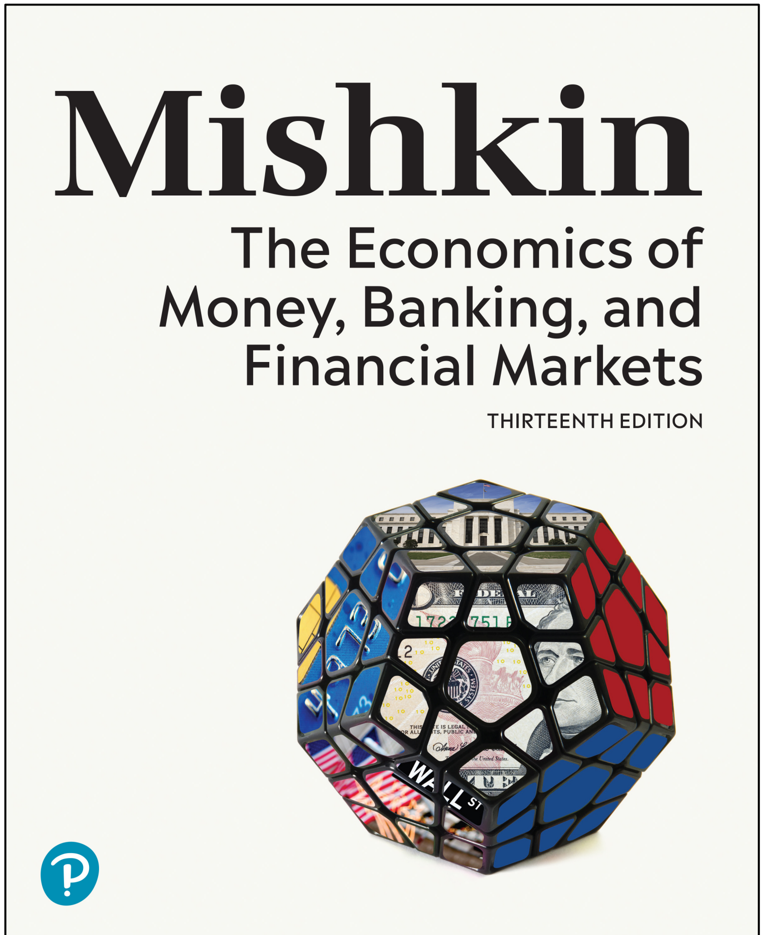
The Economics of Money, Banking, and Financial Markets

Thirteenth Edition



Chapter 13

Central Banks and the Federal Reserve System

Preview

- This chapter considers the structure and activities of central banks focusing primarily on the Federal Reserve System of the United States.

Learning Objectives (1 of 2)

13.1 Recognize the historical context of the development of the Federal Reserve System.

13.2 Describe the key features and functions of the Federal Reserve System.

13.3 Assess the degree of independence of the Federal Reserve.

13.4 Identify the ways in which the theory of bureaucratic behavior can help explain Federal Reserve actions.

Learning Objectives (2 of 2)

13.5 Summarize the arguments for and against the independence of the central bank.

13.6 Identify the similarities and distinctions in structure and independence between the European Central Bank and the Federal Reserve.

13.7 Assess the degree of independence of other major central banks around the world.

Origins of the Federal Reserve System

- **Resistance to establishment of a central bank**
  - Fear of centralized power
  - Distrust of moneyed interests
- **No lender of last resort**
  - Nationwide bank panics on a regular basis
  - Panic of 1907 so severe that the public was convinced a central bank was needed
- **Federal Reserve Act of 1913**
  - Elaborate system of checks and balances
  - Decentralized

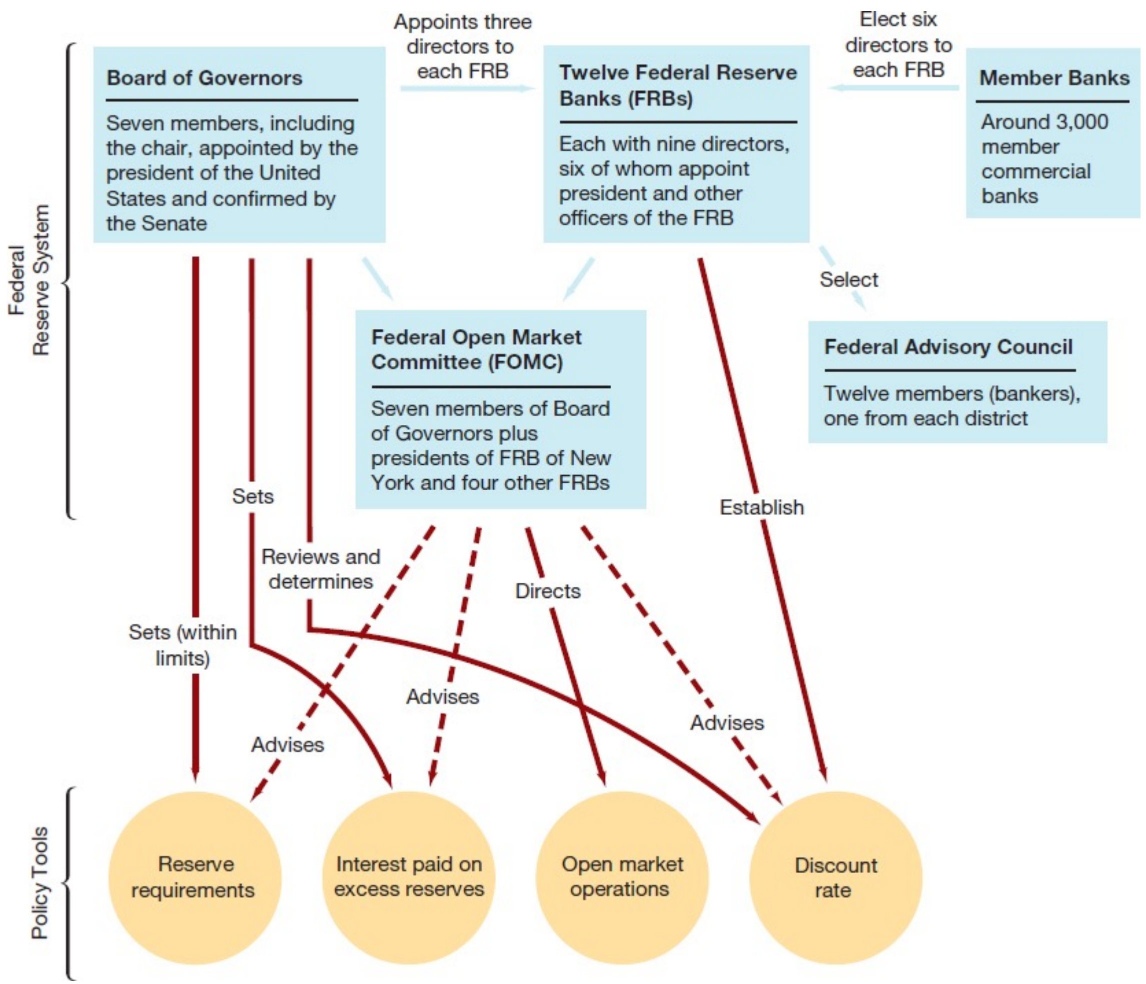
Structure of the Federal Reserve System (1 of 2)

- The writers of the Federal Reserve Act wanted to diffuse power along regional lines, between the private sector and the government, and among bankers, business people, and the public.

Structure of the Federal Reserve System (2 of 2)

- This initial diffusion of power has resulted in the evolution of the Federal Reserve System to include the following entities:
  - The Federal Reserve banks
  - The Board of Governors of the Federal Reserve System
  - The Federal Open Market Committee (FOMC)
  - The Federal Advisory Council
  - Around 1,400 member commercial banks (down from almost 2,000 a decade ago)

Figure 1 Structure and Responsibility for Policy Tools in the Federal Reserve System



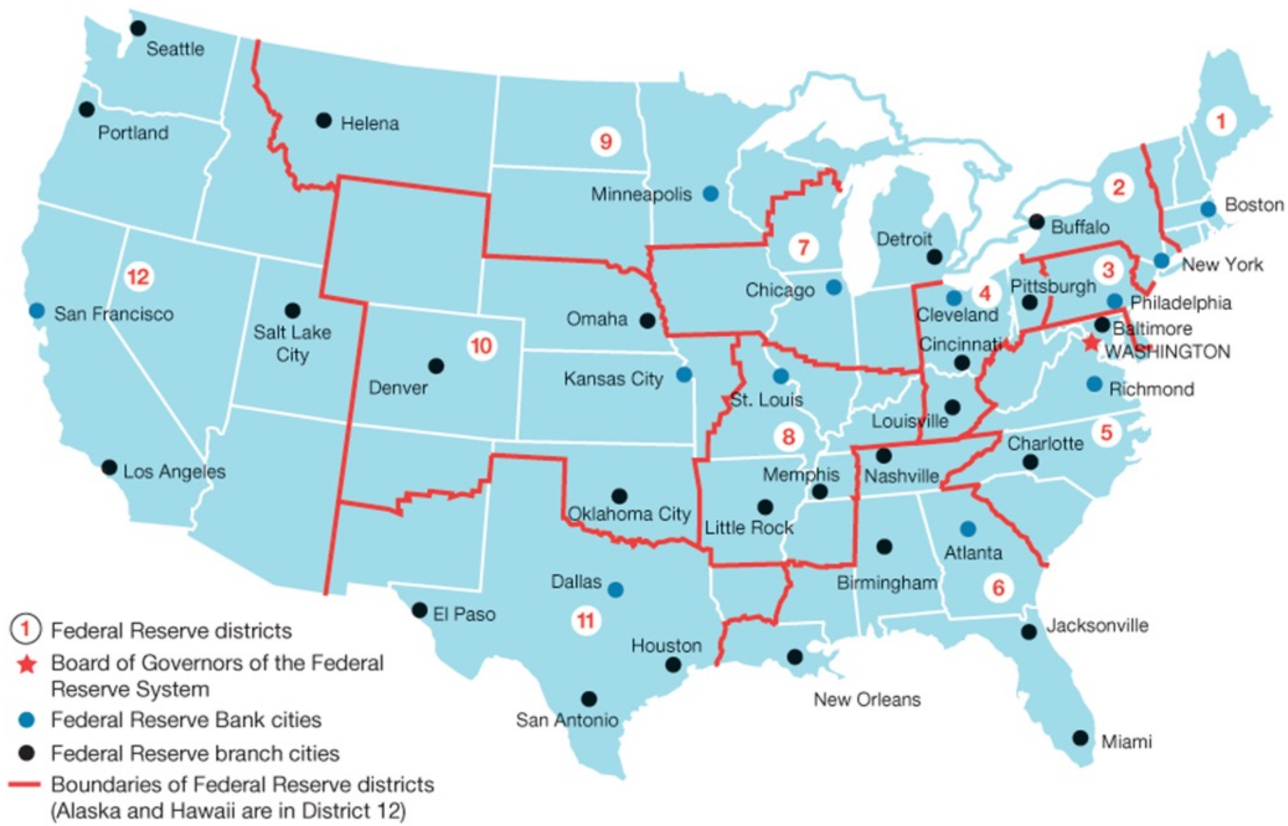
Federal Reserve Banks (1 of 2)

- Quasi-public institution owned by private commercial banks in the district that are members of the Fed system
- Every national bank was required to purchase stock in its regional Federal Reserve Bank equal to 6% of its own capital and surplus (net worth). Earn 6% dividend.
- State-chartered banks could choose to join at the start.
- On opening day, member banks transferred $241 million in reserves to the 12 regional banks, much of it in gold, to provide the system with its initial lending power.

Federal Reserve Banks (2 of 2)

- Member banks elect six directors for each district; three more are appointed by the Board of Governors
  - Three A directors are professional bankers
  - Three B directors are prominent leaders from industry, labor, agriculture, or consumer sector
  - Three C directors appointed by the Board of Governors are not allowed to be officers, employees, or stockholders of banks
  - Designed to reflect all constituencies of the public
- Nine directors appoint the president of the bank; subject to approval by Board of Governors

Figure 2 Federal Reserve System



Source: Federal Reserve Bulletin

Functions of the Federal Reserve Banks (1 of 2)

- Clear checks
- Issue new currency
- Withdraw damaged currency from circulation
- Administer and make discount loans to banks in their districts
- Evaluate proposed mergers and applications for banks to expand their activities

Functions of the Federal Reserve Banks (2 of 2)

- Act as liaisons between the business community and the Federal Reserve System
- Examine bank holding companies and state-chartered member banks
- Collect data on local business conditions
- Use staffs of professional economists to research topics related to the conduct of monetary policy

On the Special Role of the Federal Reserve Bank of New York (1 of 2)

- The Federal Reserve Bank of New York plays a special role in the Federal Reserve System for several reasons.
  - First, its district contains many of the largest commercial banks in the United States, the safety and soundness of which are paramount to the health of the U.S. financial system.
  - The second reason for the New York Fed's special role is its active involvement in the bond and foreign exchange markets.

On the Special Role of the Federal Reserve Bank of New York (2 of 2)

- The third reason for the Federal Reserve Bank of New York's prominence is that it is the only Federal Reserve Bank to be a member of the Bank for International Settlements (BIS).
- Finally, the president of the Federal Reserve Bank of New York is the only permanent voting member of the FOMC among the Federal Reserve Bank presidents, serving as the vice-chair of the committee. Thus, he or she and the chair and vice-chair of the Board of Governors are the three most important officials in the Federal Reserve System.

Gold Vault

Federal Reserve Banks and Monetary Policy

- Directors “establish” the discount rate
- Decide which banks can obtain discount loans
- Directors select one commercial banker from each district to serve on the Federal Advisory Council which consults with the Board of Governors and provides information to help conduct monetary policy
- Five of the 12 bank presidents have a vote in the Federal Open Market Committee (FOMC)

Member Banks

- Still the case that all national banks are required to be members of the Federal Reserve System
- Commercial banks chartered by states are not required but may choose to be members
- Depository Institutions Deregulation and Monetary Control Act of 1980 subjected all banks to the same reserve requirements as member banks and gave all banks access to Federal Reserve facilities

Board of Governors of the Federal Reserve System

- Seven members headquartered in Washington, D.C.
- Appointed by the president and confirmed by the Senate
- 14-year non renewable term
- Required to come from different districts
- Chairman is chosen from the governors and serves four-year term

Duties of the Board of Governors (1 of 2)

- Votes on conduct of open market operations
- Sets reserve requirements
- Controls the discount rate through “review and determination” process
- Sets margin requirements (eg 50% of purchase price for most marginable securities under Reg T, remainder borrowed from broker)
- Sets salaries of president and officers of each Federal Reserve Bank and reviews each bank’s budget

Duties of the Board of Governors (2 of 2)

- Approves bank mergers and applications for new activities.
- Specifies the permissible activities of bank holding companies.
- Supervises the activities of foreign banks operating in the United States.

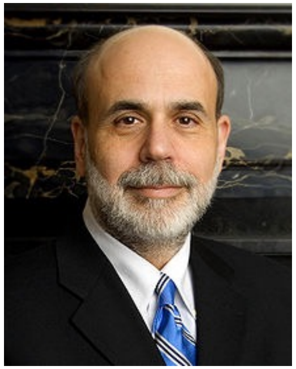
Chairman of the Board of Governors

- Advises the president on economic policy
- Testifies in Congress
- Speaks for the Federal Reserve System to the media (after each FOMC meeting and twice yearly before the House and Senate)
- May represent the United States in negotiations with foreign governments on economic matters

The Chair (1 of 4)

- Alan Greenspan took office as Chair to fill an unexpired term on August 11, 1987
- Served five terms as Chair
- Was nominated Chair by Presidents Reagan, Bush (41) and Clinton and Bush (43)

The Chair (2 of 4)



- Ben Bernanke served two terms as chair of the Federal Reserve from 2006 to 2014.
- Was member of Board of Governors before that (2002- 2005)
- Nominated in 2006 by George W. Bush (43)
- Renominated in 2010 by Barack Obama
- Before that was Economics Professor at Princeton
- Oversaw the Federal Reserve's response to the late- 2000s financial crisis.

The Chair (3 of 4)

- Janet Yellen took office as Chair on February 3, 2014
- Nominated by President Obama
- Was Vice Chair from 2010-2014

The Chair (4 of 4)

- **Jerome Powell** took office on February 5, 2018
- Nominated by President Trump and reappointed by President Biden in 2021
- Is still serving as Chair today

The Role of the Research Staff

- The Federal Reserve System is the largest employer of economists not just in the United States, but in the world.
- The most important task of the Fed's economists is to follow the incoming economic data from government agencies and private sector organizations and provide guidance to the policy-makers on the direction in which the economy might be headed and the potential impact of monetary policy actions on the economy.

Federal Open Market Committee (FOMC)

- Meets eight times a year
- Consists of seven members of the Board of Governors, the president of the Federal Reserve Bank of New York, and the presidents of four other Reserve banks
- Chairman of the Board of Governors is also chair of FOMC
- Issues directives to the trading desk at the Federal Reserve Bank of New York

Inside the Fed: The FOMC Meeting

- Report by the manager of system open market operations on foreign currency and domestic open market operations and other related issues
- Presentation of Board’s staff national economic forecast (Tealbook A)
- Outline of different scenarios for monetary policy actions (Tealbook B)
- Presentation on relevant Congressional actions
- Public announcement about the outcome of the meeting

FOMC Statement from Dec 15, 2021

(1 of 5)

- The Federal Reserve is committed to using its full range of tools to support the U.S. economy in this challenging time, thereby promoting its maximum employment and price stability goals.
- With progress on vaccinations and strong policy support, indicators of economic activity and employment have continued to strengthen. The sectors most adversely affected by the pandemic have improved in recent months but continue to be affected by COVID-19. Job gains have been solid in recent months, and the unemployment rate has declined substantially. Supply and demand imbalances related to the pandemic and the reopening of the economy have continued to contribute to elevated levels of inflation. Overall financial conditions remain accommodative, in part reflecting policy measures to support the economy and the flow of credit to U.S. households and businesses.

FOMC Statement from Dec 15, 2021
(2 of 5)

- The path of the economy continues to depend on the course of the virus. Progress on vaccinations and an easing of supply constraints are expected to support continued gains in economic activity and employment as well as a reduction in inflation. Risks to the economic outlook remain, including from new variants of the virus.

FOMC Statement from Dec 15, 2021

(3 of 5)

- The Committee seeks to achieve maximum employment and inflation at the rate of 2 percent over the longer run. In support of these goals, the Committee decided to keep the target range for the federal funds rate at 0 to 1/4 percent. With inflation having exceeded 2 percent for some time, the Committee expects it will be appropriate to maintain this target range until labor market conditions have reached levels consistent with the Committee’s assessments of maximum employment. In light of inflation developments and the further improvement in the labor market, the Committee decided to reduce the monthly pace of its net asset purchases by $20 billion for Treasury securities and $10 billion for agency mortgage-backed securities. Beginning in January, the Committee will increase its holdings of Treasury securities by at least $40 billion per month and of agency mortgage-backed securities by at least $20 billion per month. The Committee judges that similar reductions in the pace of net asset purchases will likely be appropriate each month, but it is prepared to adjust the pace of purchases if warranted by changes in the economic outlook. The Federal Reserve’s ongoing purchases and holdings of securities will continue to foster smooth market functioning and accommodative financial conditions, thereby supporting the flow of credit to households and businesses.

FOMC Statement from Dec 15, 2021
(4 of 5)

- In assessing the appropriate stance of monetary policy, the Committee will continue to monitor the implications of incoming information for the economic outlook. The Committee would be prepared to adjust the stance of monetary policy as appropriate if risks emerge that could impede the attainment of the Committee’s goals. The Committee’s assessments will take into account a wide range of information, including readings on public health, labor market conditions, inflation pressures and inflation expectations, and financial and international developments.

FOMC Statement from Dec 15, 2021
(5 of 5)

- Voting for the monetary policy action were Jerome H. Powell, Chair; John C. Williams, Vice Chair; Thomas I. Barkin; Raphael W. Bostic; Michelle W. Bowman; Lael Brainard; Richard H. Clarida; Mary C. Daly; Charles L. Evans; Randal K. Quarles; and Christopher J. Waller.

Why the Chair of the Board of Governors Really Runs the Show

- Spokesperson for the Fed and negotiates with Congress and the President
- Sets the agenda for meetings
- Speaks and votes first about monetary policy decision on Day 2
- Supervises hundreds of professional economists and advisers (contols the "intellectual engine" of the central bank)

Styles of Federal Reserve Chairs: Bernanke, Yellen and Powell Versus Greenspan

- Every Federal Reserve chair has a different style, and these styles affect how policy decisions are made at the Fed. There has been much discussion of how recent chairs of the Fed, former chairs Ben Bernanke, Janet Yellen and current chair Jerome Powell, differ from Alan Greenspan, who was the chair of the Federal Reserve Board for 19 years, from 1987 to 2006.
- Since Bernanke, Fed chair speaks last regarding economic outlook on Day 1.

How Independent Is the Fed?

- Instrument and goal independence.
- Independent revenue
- Fed’s structure is written by Congress, and is subject to change at any time.
- Presidential influence
  - Influence on Congress
  - Appoints members
  - Appoints chairman although terms are not concurrent

The Case for Independence

- Political pressure would impart an inflationary bias to monetary policy
- Political business cycle
- Could be used to facilitate Treasury financing of large budget deficits: accommodation
- Too important to leave to politicians—the principal-agent problem is worse for politicians

The Case Against Independence

- Undemocratic
- Unaccountable
- Difficult to coordinate fiscal and monetary policy
- Has not used its independence successfully

Presidential Attacks on the Independence of the Fed

- U.S. presidents, including Lyndon Johnson (D) and Richard Nixon (R) have often tried to influence Federal Reserve policies by attacking Fed independence.
- During the Clinton Administration, Treasury Secretary Robert Rubin established the doctrine that the independence of the Fed should be respected by having the president not comment on Fed policies—which was then adopted by George W. Bush and later by Barack Obama.
- Donald Trump was more vocal than recent predecessors

Explaining Central Bank Behavior (1 of 2)

- One view of government bureaucratic behavior is that bureaucracies serve the public interest (this is the **public interest view**). Yet some economists have developed a theory of bureaucratic behavior that suggests other factors that influence how bureaucracies operate.
- The theory of bureaucratic behavior may be a useful guide to predict what motivates the Fed and other central banks.

Explaining Central Bank Behavior (2 of 2)

- Theory of bureaucratic behavior: the objective of a bureaucracy is to maximize its own welfare that is related to power and prestige
  - Fight vigorously to preserve autonomy
  - Avoid conflict with more powerful groups
- Does not rule out altruism

Inside the Fed: The Evolution of the Fed's Communication Strategy

- The Fed has dramatically increased its transparency in recent years
  - In 2011, Ben Bernanke began a policy where following the January, April, June, and November FOMC meetings, the Chair of the Board of Governors holds a press conference to clarify monetary policy communications.
  - In 2019, Jerome Powell expanded the schedule to include a press conference after every scheduled meeting (eight times per year).
  - Greater transparency has been provided with the announcement of a specific numerical target for the inflation rate.

Structure and Independence of the European Central Bank

- Patterned after the Federal Reserve
- Central banks from each country play similar role as Fed banks
- Executive Board:
  - President, vice-president, and four other members
  - Eight year, nonrenewable terms
- Governing Council

Differences Between the European System of Central Banks and the Federal Reserve System

- National Central Banks control their own budgets and the budget of the ECB
- Monetary operations are not centralized
- Does not supervise and regulate financial institutions

Governing Council

- Monthly meetings at ECB in Frankfurt, Germany
- Nineteen National Central Bank heads and six Executive Board members
- Operates by consensus
- ECB announces the target rate and takes questions from the media
- To stay at a manageable size as new countries join, the Governing Council will be on a system of rotation.

How Independent Is the ECB?

- Most independent in the world
- Members of the Executive Board have long terms
- Determines own budget
- Less goal independent
  - Price stability
- Charter cannot by changed by legislation; only by revision of the Maastricht Treaty

Structure and Independence of Other Foreign Central Banks

- Bank of Canada
  - Essentially controls monetary policy
- Bank of England
  - Has some instrument independence.
- Bank of Japan
  - Recently (1998) gained more independence (Bank of Japan Act of 1997)